

Review Report: MaskClass — A Masked Denoising Autoencoder for Scalable and Accurate Single-Cell RNA-seq Analysis

Target Venue: NeurIPS 2026

Review Date: March 20, 2026

CRITICAL FORMATTING ISSUE

The paper footer reads: "Submitted to the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026)."

If the target venue is NeurIPS 2026, the template, formatting, and page limits must be changed to the NeurIPS 2026 LaTeX style. This is a desk-reject-level issue if submitted as-is.

Task 1: Correctness Audit

1A: Mathematical Verification

#	Item	Location	Step-by-Step Verification	Correct?	Issue and Fix
1	Gene scaling (Winsorize → Standardize → Rescale)	Sec. 3.3	Winsorize clips to $[q^{(2)}, q^{(99.5)}]$. Standardize subtracts mean and divides by σ . Rescale maps $[z_{\min}, z_{\max}] \rightarrow [-1, 1]$. Steps are individually correct and monotone. Guard $\epsilon > 0$ prevents division by zero.	Yes	No issue. However, the paper does not state what happens when $\sigma_j = 0$ (constant gene after winsorizing). The $\max(\sigma_j, \epsilon)$ handles it numerically but the output is arbitrary. Consider documenting or filtering these genes.
2	Masked corruption — Bernoulli masks (Eq. in Sec. 3.4)	Sec. 3.4	$m^{(0)}_{ij} \sim \text{Bernoulli}(\pi^{(0)}_{ij})$ with $\pi^{(0)}_{ij} = p_{\text{zero}} \cdot p_{\text{bio}}(i,j)$. For observed zeros, $p_{\text{bio}} \in [0,1]$ so $\pi^{(0)}_{ij} \in [0, p_{\text{zero}}] \subset [0,1]$. Valid	Yes (with caveat)	The equation for the zero mask is cut off mid-line on page 4: " $m^{(0)}_{ij} \sim \text{Bernoulli}(\pi^{(0)}_{ij})$, $\pi^{(0)}_{ij} = p_{\text{zero}} p_{\text{bio}}(i,j)$ for $(i,j) \in Z, m$ " — the trailing "m" is a typographical artifact. FIX: Remove the dangling "m"

Bernoulli parameter.
For non-zeros,
 $m^{(nz)}_{ij} \sim$
Bernoulli(p_{nz}). Fine.

3	Noise for masked zeros	Sec. 3.4	$\eta_{ij} \sim \text{Uniform}(0, v)$, $v = 0.02$. Then $\tilde{x}_{ij} = \log_2(1 + \eta_{ij} \cdot c^{\max_j})$. Since $\eta_{ij} \in [0, 0.02]$ and $c^{\max_j} \geq 0$, the argument of $\log_2$ is ≥ 1 , so $\tilde{x}_{ij} \geq 0$. Dimensionally consistent (log-space value).	Yes	No issue.
4	Weighted masked MSE loss L_{mask}	Sec. 3.4	Numerator: $w_{\text{bio}} \sum_{\{Z\}} m^{(0)}_{ij} r^2_{ij} + w_{\text{nz}} \sum_{\{N\}} m^{(nz)}_{ij} r^2_{ij}$. Denominator: $w_{\text{bio}} \sum_{\{Z\}} m^{(0)}_{ij} + w_{\text{nz}} \sum_{\{N\}} m^{(nz)}_{ij}$. This is a weighted average of squared residuals over masked entries. Both weights are 2.0.	Yes (with caveat)	Since $w_{\text{bio}} = w_{\text{nz}} = 2.0$, they cancel out. The denominator counts total masked entries ($\times 2.0$). The formulation is correct but redundant — having equal weights is equivalent to an unweighted average over all masked entries. This undermines the claim that the scheme differentially weights zeros vs. non-zeros. The <i>differential treatment</i> comes from different masking rates, not different loss weights. Clarify in text or use different weight values.
5	Biozero regularizer L_{bio}	Sec. 3.4	$L_{\text{bio}} = \sum_{\{Z\}} p_{\text{bio}}(i,j) (\hat{s}_{ij} - s0_j)^2 / \sum_{\{Z\}} p_{\text{bio}}(i,j)$. This is a p_{bio} -weighted MSE pushing reconstructions toward the scaled zero for likely biozeros. Correct as a regularizer.	Yes	No issue with the math. However, $s0_j = [\text{scale}(0)]_j$ is never explicitly defined. It should be stated that $\text{scale}(0)$ applies the winsorize-standardize-rescale pipeline to the value 0. This is potentially ambiguous: does it use the gene-specific statistics from the full data? Presumably yes, but state

this explicitly.

6	Non-zero reconstruction L_{nz}	Sec. 3.4	$L_{nz} = \sum_{\{N\}} r^2_{ij} /$	N	. Standard MSE over non-zero entries. Correct.
7	Final objective $L = L_{mask} + \lambda_{bio} L_{bio} + \lambda_{nz} L_{nz}$	Sec. 3.4	Linear combination with $(\lambda_{bio}, \lambda_{nz}) = (2.0, 0.75)$.	Yes	No mathematical issue, but note : L_{nz} is computed over ALL non-zero entries (not just masked ones), while L_{mask} only covers masked entries. This means the model receives direct supervision on unmasked non-zeros through L_{nz} . This is fine methodologically but means the "masked" objective is supplemented by a full-matrix reconstruction term on non-zeros. This should be discussed more clearly.
8	Biological zero probability $p_{bio}(i,j)$	Sec. 3.5	Bayes' rule application: $P(t_{ij}=0)$	$c_{ij}=0) = P(c_{ij}=0)$	$t_{ij}=0)P(t_{ij}=0) / P(c_{ij}=0)$. With $c_{ij} = (1-d_{ij})t_{ij}$: $P(c_{ij}=0) = \delta_{ij} + (1-\delta_{ij})f_{ij}(0)$. $P(c_{ij}=0$
9	Reconstruction-informed shrinkage	Sec. 3.7	$\hat{u}_{ij} = 2^{(\hat{x}_{ij})} - 1$. $w_{ij} = 1/(1 + \hat{u}_{ij}/\max(\tilde{u}_j, \epsilon))$. $\hat{u}^{(shrink)}_{ij} = \hat{u}_{ij}(1 - \alpha w_{ij})$. When $\hat{u}_{ij}$ is small relative to $\tilde{u}_j$, $w_{ij} \rightarrow 1/(1+0) = 1$, so shrinkage is maximal: $\hat{u}^{(shrink)} = \hat{u}_{ij}(1-\alpha)$. When $\hat{u}_{ij} \gg \tilde{u}_j$, $w_{ij} \rightarrow 0$ and shrinkage vanishes. Logically correct.	Yes (with caveat)	The formula $\hat{u}_{ij} = 2^{(\hat{x}_{ij})} - 1$ inverts $x = \log_2(1+u)$. This is correct only if $\hat{x}_{ij}$ is in $\log_2(1+CP10k)$ space. But $\hat{x}_{ij} = [\text{invscale}(\hat{S})]_{ij}$ is in log-normalized space. If invscale correctly undoes scale, this is fine. However , the paper says "reconstructed CP10k-normalized count scale corresponding to $\hat{x}_{ij}$." This phrasing is confusing — $\hat{x}_{ij}$ is already in log space; $\hat{u}_{ij}$ is in CP10k-normalized count space. Clarify.
10	MSE/MAE metric definitions	Sec. 3.8	$MSE = (1/NG) \sum (x^*_{ij} - x'_{ij})^2$. Standard. MAE analogous.	Yes	No issue.

Subset metrics
restrict the average to
the subset.

11	gNRMSE definition	Sec. 3.8	$gNRMSE = (1/G) \sum_j [RMSE_j / \max(sd(x^*_j), \epsilon)]$. Per-gene normalized RMSE, averaged over genes.	Yes	No issue mathematically. However, $gNRMSE > 1$ (MaskClass gets 1.17) means reconstruction error exceeds gene-level standard deviation on average. This is not highlighted or discussed.
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Summary of Mathematical Issues

- **1 Error (Medium-High):** Biological zero probability formula (Sec. 3.5) has an incorrect numerator. Fix: numerator should be $f_{ij}(0)$, not $(1-\delta_{ij})f_{ij}(0)$.
- **1 Typographical bug:** Dangling "m" in the masking equation (Sec. 3.4).
- **Multiple clarity issues:** Equal weights $w_{bio} = w_{nz}$ not discussed; $s0_j$ definition implicit; shrinkage variable naming confusing; $gNRMSE > 1$ not discussed.

1B: Notation Consistency Table

Symbol	First Defined	Meaning	Used Consistently?	Conflict
X_{log}	Sec. 3.1 (Notation)	Observed log-normalized matrix	Yes	—
X^*_{log}	Sec. 3.1 (Notation)	Ground-truth log-normalized matrix	Yes	—
X'_{log}	Sec. 3.1 (Notation)	Reconstructed logcounts (final output)	Yes	—
$\hat{X}_{log}$	Sec. 3.7 / Alg. 1	Reconstructed logcounts (before shrinkage)	Partially	$\hat{X}_{log}$ vs X'_{log} distinction only clear in Sec. 3.7. In Sec. 3.2 it says " $X'_{log} = \text{invscale}(\hat{S})$ " which should be $\hat{X}_{log}$.
S	Sec. 3.2 / 3.3	Scaled input matrix	Yes	—

$\hat{S}$	Sec. 3.2 / 3.4	Scaled reconstruction matrix	Yes	—
s_{ij}	Sec. 3.4	Entry of scaled matrix	Yes	—
$\tilde{s}_{ij}$	Sec. 3.4	Corrupted scaled input	Yes	—
$\hat{s}_{ij}$	Sec. 3.4	Reconstructed scaled entry	Yes	—
C / c_{ij}	Sec. 3.1 (Notation)	Count matrix / entry	Yes	—
r_{ij}	Sec. 3.4	Residual $\hat{s}_{ij} - s_{ij}$	Yes	—
$p_{\text{bio}}(i,j)$	Sec. 3.4 / 3.5	Biozero probability	Yes	—
$p_{\text{zero}}, p_{\text{nz}}$	Sec. 3.4	Masking rates	Yes	—
$m^{(0)}_{ij}$	Sec. 3.4	Zero mask	Yes	—
$m^{(\text{nz})}_{ij}$	Sec. 3.4	Non-zero mask	Yes	—
Z, N	Sec. 3.4	Zero/non-zero entry sets	Partially	Z is also used for standardized values z_{ij} in Sec. 3.3. Lowercase z_{ij} vs. set Z — could confuse.
d	Sec. 3.2	Bottleneck dimension	Yes	Also d_{ij} is dropout indicator in Sec. 3.5. Conflict.
f_{θ}, g_{ϕ}	Sec. 3.2	Encoder / decoder	Yes	—
δ_{ij}	Sec. 3.5	Dropout probability	Yes	—
ε	Sec. 3.3	Numerical stability constant	Yes	—
α	Sec. 3.7	Shrinkage strength	Yes	—
$\hat{u}_{ij}$	Sec. 3.7	Reconstructed	Yes	—

$\tilde{u}_j$	Sec. 3.7	Robust reference level for gene j	Yes	—
w_{ij}	Sec. 3.7	Shrinkage weight	Partially	w_{ij} in Sec. 3.7 (shrinkage weight) vs. w_{bio} , w_{nz} in Sec. 3.4 (loss weights). Same letter "w" for different purposes.
B	Multiple	Biozero set (Sec. 3.8) AND mini-batch (Sec. 3.4)	No	Conflict: B = biozero entries in Sec. 3.8 metrics, B = mini-batch in Sec. 3.4.

Notation Issues Summary

- **d** used for both bottleneck dimension (Sec. 3.2) and dropout indicator d_{ij} (Sec. 3.5)
- **B** used for both mini-batch (Sec. 3.4) and biozero entry set (Sec. 3.8)
- **w** used for both shrinkage weights (Sec. 3.7) and loss weights (Sec. 3.4)
- $X'_{\log}$ vs. $\hat{X}_{\log}$ distinction introduced late and inconsistently applied

1C: Reference Integrity

- ✓ Every `\cite{}` appears to have a corresponding bibliography entry (18 references, all present)
- ✓ Bibliography entries all appear to be cited
- **Figure 1 caption mentions " $p^{(0)}_{bio(i,j)}$ " with superscript (0) — this superscript does not appear anywhere in the main text.** The text uses $p_{bio(i,j)}$ without a superscript. Inconsistency.
- **"Algorithm 1 (Algorithm 1)" — redundant self-reference in Sec. 3.9.**
- ✓ Table 1 and Table 2 are correctly referenced
- ✓ Figure 2 and Figure 3 are correctly referenced
- **No explicit reference to which figure shows which scenario breakdown** — Tables 1 and 2 show averages but per-scenario breakdowns are not provided in any table or supplementary.
- *The paper references "DEFacGroup row-data columns" (Sec. 3.1) without formally defining what these are** — assumes reader familiarity with Splatter internals.
- ✓ Cross-references between sections (e.g., "Section 3.4", "Section 3.5") appear correct.
- **No supplementary material is included or referenced**, despite the paper mentioning versioned artifact packages in Sec. 5.

Broken/Suspicious References

- 1. Figure 1 caption: superscript (0) on p_bio not consistent with main text
- 2. Sec. 3.9: "Algorithm 1 (Algorithm 1)" — duplicate reference
- 3. No per-scenario results table despite claiming "wins on 12/13 scenarios" — reader cannot verify

Task 2: Anti-Hallucination Sweep

Since I only have the single version of this paper (no separate "original draft" vs. "rewritten paper"), I will verify internal consistency of all quantitative claims against the tables and text within the paper itself.

#	Claim in Paper	Source Within Paper	Status
1	"MaskClass achieves MSE = 0.226"	Table 1: MSE = 0.2265	Grounded (rounded)
2	"Dropout-MSE = 0.308"	Table 1: Dropout-MSE = 0.3080	Grounded
3	"Biozero-MSE = 0.0117"	Table 1: 0.0117	Grounded
4	"Reduces overall MSE by 37% relative to DCA"	DCA MSE = 0.3598. (0.3598 - 0.2265)/0.3598 = 37.0%	Grounded
5	"Dropout-MSE by 34% relative to DCA"	DCA = 0.4652. (0.4652-0.3080)/0.4652 = 33.8% ≈ 34%	Grounded (rounded)
6	"Biozero-MSE by 29% relative to DCA"	DCA = 0.0164. (0.0164-0.0117)/0.0164 = 28.7% ≈ 29%	Grounded (rounded)
7	"Wins on 12/13 scenarios for overall MSE"	No per-scenario table provided	UNVERIFIABLE — reader cannot check this claim
8	"G = 1000 genes"	Sec. 3.1	Grounded (simulation parameter)
9	"N = 1000 (tuning), N = 5000 (test)"	Sec. 3.1	Grounded
10	"N ∈ {10000, ..., 100000} (scaling)"	Sec. 3.1	Grounded (matches Figure 3

11	"13 scenarios"	Sec. 3.1 — lists 1 baseline + 6 families $\times$ 2 variants = 13	Grounded
12	"de.prob = 0.10 in all grouped scenarios"	Sec. 3.1	Grounded (stated)
13	"5 repeats with different random seeds"	Sec. 4.1	Grounded (stated)
14	"(pzero, pnz) = (0.01, 0.05)"	Sec. 3.4	Grounded
15	"(wbio, wnz) = (2.0, 2.0)"	Sec. 3.4	Grounded
16	"(λ bio, λ nz) = (2.0, 0.75)"	Sec. 3.4	Grounded
17	"Bottleneck dimension d = 16"	Sec. 3.2	Grounded
18	"Encoder widths 128 $\rightarrow$ 64 $\rightarrow$ d, decoder d $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ G"	Sec. 3.2	Grounded
19	" α = 0.3"	Sec. 3.7	Grounded
20	"v = 0.02"	Sec. 3.4	Grounded
21	"400 epochs"	Sec. 3.6	Grounded
22	"Learning rate 10^{-4} , weight decay 10^{-4} "	Sec. 3.6	Grounded
23	"Batch size 32"	Sec. 3.6	Grounded
24	"Biozero errors are lowest for conservative strategies (Baseline, SAVER)"	Table 1: Baseline Biozero-MSE=0.0000, SAVER=0.0020	Grounded
25	Abstract: "applicable to other high-dimensional, sparse, and noisy data modalities including other single-cell omics proteomics, spatial transcriptomics"	No experiments on any of these modalities	OVERCLAIM — not tested, should say "potentially applicable"
26	"Benchmarking against state-of-the-art denoising methods"	Baselines are DCA (2019), MAGIC (2018), AutoClass (2022), ccImpute (2022), SAVER (2018)	MISLEADING — several baselines are 5+ years old. Missing scVI, scGEN, ALRA, and more recent methods.

Anti-Hallucination Summary

- **Total claims checked:** 26
 - **Grounded:** 21
 - **Needs verification / Unverifiable:** 2 (claim #7: 12/13 win claim; claim #26: "state-of-the-art" characterization)
 - **Overclaim:** 1 (claim #25: applicability to other modalities without evidence)
 - **Misleading:** 1 (claim #26: baselines are not current state-of-the-art)
 - **Likely hallucinated:** 0
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Task 3: Simulated Peer Review

REVIEWER 1: The Senior Expert

Perspective: Deep technical knowledge of scRNA-seq computational methods. Published 50+ papers including foundational work on measurement models and imputation. Knows the zero-inflation debate intimately.

Summary of the Paper

This paper proposes MaskClass, a masked denoising autoencoder for scRNA-seq data that distinguishes observed zeros from non-zeros through differential masking, biozero-aware regularization, and post-hoc shrinkage. The method is evaluated exclusively on synthetic data generated by Splatter and compared against several baselines.

Summary of Strengths

1. **Well-motivated zero-aware design (Sec. 3.4–3.5):** The differential treatment of zeros and non-zeros during masking, combined with biozero probability estimation, is a principled approach to the fundamental ambiguity of zeros in scRNA-seq. This is the paper's strongest conceptual contribution.
2. **Comprehensive synthetic benchmark (Sec. 3.1):** The 13-scenario design space covering balanced/imbalanced groups, rare populations, batch effects, and trajectories with both experiment-level and group-level dropout is thorough for synthetic evaluation.
3. **Honest biozero trade-off discussion (Sec. 4.1):** The paper acknowledges that MaskClass trades higher biozero-MSE for better dropout recovery, which is refreshingly candid compared to papers that only highlight wins.
4. **Reproducible design:** Fixed seeds, detailed hyperparameters, open-source code, and

4. **Reproducible design:** Fixed seeds, detailed hyperparameters, open-source code, and

deterministic synthetic data generation support reproducibility.

Summary of Weaknesses

1. **No real data experiments (CRITICAL):** The paper evaluates exclusively on Splatter-simulated data. Splatter generates data from a parametric model (Gamma-Poisson with logistic dropout), and the biozero probability estimation in Sec. 3.5 uses a remarkably similar negative-binomial dropout model. This creates a **circularity concern**: the method's assumptions closely match the data-generating process, inflating reported gains. No NeurIPS reviewer will accept a methods paper in 2026 that lacks real-data validation. Standard benchmarks (e.g., Zheng PBMC 68k, Baron pancreas, Zeisel brain) are readily available.
2. **Missing critical baselines:** scVI (Lopez et al., 2018), ALRA (Linderman et al., 2022 — already cited!), and scGEN are not compared. ALRA is particularly relevant as it is also a zero-preserving imputation method by authors the paper already cites. The baselines range from 2017 to 2022; no 2023–2025 methods are included, which is unacceptable for a 2026 venue.
3. **No ablation study of the three proposed components:** The paper introduces (i) differential masking, (ii) biozero regularization, and (iii) post-hoc shrinkage as three contributions. There is no ablation removing each component individually. Without this, the reader cannot determine which component drives the gains.
4. **Mathematical error in p_{bio} (Sec. 3.5):** The Bayesian derivation has an incorrect numerator. When $t_{ij} = 0$, the observed count is zero regardless of whether dropout occurs, so $P(c_{ij}=0, t_{ij}=0) = f_{ij}(0)$, not $(1-\delta_{ij})f_{ij}(0)$. This underestimates biological zero probability.

Questions for Authors

1. How does MaskClass perform on real scRNA-seq datasets (e.g., PBMC, pancreas, or cortex datasets) where ground truth is not available? Can you report clustering metrics (ARI, NMI) or differential expression concordance?
2. Why is ALRA (Linderman et al., 2022) not included as a baseline, given that it is a zero-preserving method and the paper already cites Linderman et al.?
3. What is the ablation performance when removing (a) biozero regularization, (b) the shrinkage step, and (c) differential masking rates? Which component matters most?

Missing References

- Lopez et al. (2018), "Deep generative modeling for single-cell transcriptomics" (scVI) — the most widely used deep learning baseline
- Linderman et al. (2022), ALRA — already cited but not compared against
- Gayoso et al. (2022), scvi-tools
- Xu et al. (2021), scIGANs
- Recent transformer-based methods (scBERT, Geneformer, scGPT) for positioning

Minor Issues

- Sec. 3.4: Dangling "m" at the end of the masking equation
- Sec. 3.9: "Algorithm 1 (Algorithm 1)" — redundant
- Figure 1 caption uses $p^{(0)}_{\text{bio}}$ superscript not found in main text
- The training objective in Figure 1 header shows " $L = L_{\text{mask}} + L_{\text{bio}}$ " but omits L_{nz}

Scores

- Soundness: 2 / 4
- Contribution/Significance: 2 / 4
- Novelty: 2 / 4
- Clarity: 3 / 4
- Reproducibility: 4 / 5
- Overall: 4 / 10
- Confidence: 5 / 5

Recommendation: Reject

REVIEWER 2: The Empiricist

Perspective: Experimental rigor specialist. Scrutinizes every table, baseline, statistical test, and reproducibility claim. Cares about practical impact.

Summary of the Paper

MaskClass combines masked autoencoding with biozero-aware regularization and post-hoc shrinkage for scRNA-seq denoising. It is benchmarked against six methods on 13 synthetic scenarios, showing improvements in overall and dropout-specific reconstruction metrics.

Summary of Strengths

1. **Clean experimental protocol (Sec. 3.1):** Separate tuning/test/scaling splits with fixed seeds. Five random repeats per method per scenario. This is above-average experimental hygiene for the subfield.
2. **Multiple granular metrics (Sec. 3.8):** Reporting MSE on dropout-specific, biozero-specific, and marker-gene subsets is much more informative than overall MSE alone. This reveals the biozero trade-off clearly.
3. **Runtime scaling analysis (Fig. 3):** Including runtime comparisons is valuable for practical deployment, especially as single-cell datasets grow to millions of cells.
4. **Open-source code availability:** Source code at GitHub link supports independent verification.

Summary of Weaknesses

1. **Synthetic-only evaluation is disqualifying (CRITICAL):** Real biological data has structure that Splatter cannot capture: doublets, ambient RNA, cell-cycle effects, spatial autocorrelation in droplet capture, etc. Without real-data benchmarks, the practical value is unknown. At minimum: (a) PBMC or similar with cluster-level ARI/NMI before/after imputation, (b) gene-gene correlation recovery, (c) differential expression analysis comparison.
2. **No statistical significance tests:** Table 1 reports means over 5 seeds but no standard deviations or confidence intervals for any method (only MaskClass biozero-MSE std is shown in Figure 2). A 37% MSE reduction could be within noise. Without error bars in tables or paired statistical tests across scenarios, the gains are not statistically established.
3. **Runtime: MaskClass is the second-slowest method (Fig. 3):** At 100k cells, MaskClass takes ~2600s vs. DCA at ~1600s and MAGIC at ~2100s. Only AutoClass is slower. The paper does not discuss this weakness. Scalability claims in the abstract are undermined.
4. **No downstream task evaluation:** Denoising is not an end in itself; it serves clustering, trajectory inference, differential expression, etc. Without showing that MaskClass improves downstream analyses, the practical significance is unclear.
5. **Per-scenario results hidden:** The paper claims "wins on 12/13 scenarios" but provides no per-scenario breakdown. Table 1 only shows scenario-averaged values. This prevents the reader from identifying failure modes.
6. **No hyperparameter sensitivity analysis:** The method has at least 7 hyperparameters (p_{zero} , p_{nz} , v , α , λ_{bio} , λ_{nz} , d). All are fixed and never varied. How sensitive is performance to these choices?

Questions for Authors

1. Can you provide per-scenario MSE results (as a supplementary table) so reviewers can verify the 12/13 win claim?
2. What are the standard deviations across seeds for all methods in Table 1?
3. How does MaskClass perform when hyperparameters are not optimally tuned? What is the sensitivity to α (shrinkage) and λ_{bio} (biozero regularization weight)?
4. What is the GPU hardware used for runtime comparison? Is it the same for all methods?

Missing References

- Benchmark studies: Hou et al. (2020) is cited but more recent comprehensive benchmarks from 2022–2025 should be discussed.
- Any real-data benchmark paper from the scRNA-seq imputation literature.

Minor Issues

- Table 1: "Baseline" is presumably the observed data without imputation — this should be explicitly stated in the table caption or text, not left implicit.
- Figure 2: y-axis label for Biozero-MSE panel starts at -0.02, which is impossible for MSE. This appears to be a matplotlib default axis issue. Fix the y-axis lower bound to 0.
- Figure 3: No error bars for runtime (only single runs?). State this explicitly.
- The paper claims "scalable" in the title but MaskClass is among the slowest methods in Figure 3.

Scores

- Soundness: 2 / 4
- Contribution/Significance: 1 / 4
- Novelty: 2 / 4
- Clarity: 3 / 4
- Reproducibility: 4 / 5
- Overall: 3 / 10
- Confidence: 4 / 5

Recommendation: Reject

REVIEWER 3: The Adversarial Skeptic

Perspective: Actively looks for reasons to reject. Questions every claim. Tired, skimming, and writing at 11 PM before the deadline.

Summary of the Paper

Yet another autoencoder for scRNA-seq denoising, this time with a masking strategy. Evaluated only on synthetic data. I am not convinced this advances the field.

Summary of Strengths

1. **The biozero-aware masking idea (Sec. 3.4) is interesting in principle.** Differentiating corruption strategies for zeros vs. non-zeros is a reasonable idea for this domain.
2. **The paper is well-written and the methods section is detailed.** I could reimplement this from the description.
3. **The code is open source.** Good.

Summary of Weaknesses

1. **I am not convinced this is a NeurIPS paper.** The novelty is a combination of (a) masked autoencoders (well-established), (b) a biozero prior from standard NB modeling (well-established in SAVER, scImpute, etc.), and (c) a simple post-hoc shrinkage heuristic. None of these components are individually novel. The combination may be useful engineering, but I struggle to identify a conceptual or theoretical contribution that would excite the NeurIPS audience.
2. **Synthetic-only evaluation is a fatal flaw.** I have reviewed dozens of scRNA-seq methods papers. Every single one that got accepted at a top venue included real-data experiments. The Splatter simulator uses the same parametric family (NB + logistic dropout) as the method's own biozero estimation (Sec. 3.5). This is essentially evaluating a model on data that matches its own assumptions. I would be shocked if the gains hold on real data.
3. **The shrinkage mechanism (Sec. 3.7) is ad hoc.** Why $\alpha = 0.3$? Why a sigmoidal weight w_{ij} with that specific functional form? There is no theoretical justification, no ablation over α , and no comparison to simpler alternatives (e.g., soft-thresholding, wavelet shrinkage).
4. **Overclaiming in the abstract:** "significantly improves data quality" — significance is not established (no p-values, no downstream impact). "Scalable" — the method is the second-slowest in Figure 3. "Applicable to other high-dimensional, sparse, and noisy data modalities" — zero evidence for this.
5. **Self-citation bias:** ccImpute [Malec et al., 2022] is the authors' prior work and is included as

a baseline. It performs poorly (worst dropout-MSE). This creates an unfavorable comparison that may inflate MaskClass's apparent contribution. Meanwhile, much more competitive recent methods are absent.

6. **gNRMSE > 1 is not discussed.** MaskClass achieves gNRMSE = 1.17, meaning per-gene reconstruction error exceeds the gene's own standard deviation on average. This is a concerning result that suggests the method is not performing well in absolute terms, only relatively better than weak baselines.
7. **No error analysis or failure cases.** On which scenario does MaskClass lose (the 1/13 case)? What types of genes or cells cause the largest errors? What does the latent space look like? There is no qualitative analysis whatsoever.

Questions for Authors

1. Why should I believe results on Splatter-simulated data will transfer to real biological data, given that your biozero estimation model mirrors Splatter's data-generating process?
2. What is the one scenario where MaskClass does not win? Why?
3. If I remove the biozero regularization and the shrinkage step, how much of the gain remains (i.e., what fraction comes from the masking strategy alone)?
4. Why not compare against ALRA, which is explicitly designed for zero-preserving imputation and is by authors you already cite?

Missing References

- ALRA (Linderman et al., 2022) — should be a baseline, not just a citation
- scVI (Lopez et al., 2018) — most widely used deep learning method
- scSphere (Ding and Regev, 2021) — geometric deep learning approach
- Recent foundation model approaches (scGPT, Geneformer) for context

Minor Issues

- Title says "Scalable" but Figure 3 contradicts this — MaskClass is slower than DCA and SAVER at scale
- The abstract is too long and buries the actual contributions
- No broader impact section beyond a brief ethical disclaimer

Scores

- Soundness: 2 / 4

- Contribution/Significance: 1 / 4
- Novelty: 1 / 4
- Clarity: 3 / 4
- Reproducibility: 4 / 5
- Overall: 3 / 10
- Confidence: 4 / 5

Recommendation: Reject

META-REVIEWER (Area Chair) Synthesis

Consensus Strengths

1. **Principled zero-aware design:** All three reviewers recognize the differential masking + biozero regularization as a well-motivated idea.
2. **Thorough synthetic evaluation protocol:** Fixed seeds, multiple scenarios, five repeats, separate splits.
3. **Good reproducibility practices:** Open source, detailed hyperparameters, deterministic data generation.

Consensus Weaknesses

1. **No real-data experiments (unanimous, CRITICAL):** All three reviewers flag this as disqualifying. The circularity between the NB-based biozero estimation and Splatter's NB data-generating process is a particularly damning observation.
2. **Missing modern baselines (unanimous):** scVI, ALRA, and recent methods from 2023–2025 are absent. The comparison set is dominated by methods from 2018–2019.
3. **No ablation study (unanimous):** Three proposed components but no systematic ablation to attribute gains.
4. **No downstream task evaluation (R2, R3):** Denoising metrics alone do not establish practical value.
5. **Overclaiming (R1, R3):** "Scalable" contradicted by Figure 3; "significantly improves" without significance tests; cross-modality claims without evidence.

Points of Disagreement Between Reviewers

- **Novelty:** R1 sees the zero-aware design as moderately interesting; R3 considers it incremental engineering. R2 is closest to the likely NeurIPS reviewer consensus – the

incremental engineering. R3 is closer to the likely NeurIPS reviewer consensus — the components individually exist, and their combination, while sensible, lacks theoretical depth.

- **Mathematical error severity:** R1 flags the `p_bio` formula error as medium-high; R2 and R3 do not discuss it in depth. R1 is correct that this is a meaningful error but it likely does not catastrophically affect results since the impact is a modest underestimation of `p_bio`.

Likely Outcome at This Venue

- **Current verdict:** Reject
- **Confidence in this verdict:** High

What Would Flip a Reject to Borderline Accept

1. **Add real-data experiments on at least 3 standard benchmarks** (e.g., PBMC 68k, Baron pancreas, Zeisel brain) with downstream metrics (ARI, NMI, DE concordance, gene-gene correlation preservation).
2. **Add modern baselines** (scVI, ALRA at minimum; ideally one 2024–2025 method).
3. **Add a systematic ablation study** of the three proposed components.

What Would Flip a Borderline Accept to Strong Accept

1. **Theoretical analysis** of when/why the biozero-aware masking provably helps (e.g., sample complexity bounds, recovery guarantees under specific noise models).
2. **Large-scale real-data demonstration** on a million-cell atlas dataset showing practical scalability and downstream biological discovery.

Task 4: Addressing Every Review Concern

#	Concern	Reviewer	Severity	Addressed in Paper?	Where	How to Fix if Not Addressed
1	No real-data experiments	R1, R2, R3	Critical	No	N/A	Run on PBMC 68k, Baron pancreas, Zeisel brain. Report ARI, NMI for clustering; DE concordance; gene-gene correlation.
2	Missing	R1, R3	Critical	No	N/A	Install scvi-tools, run scVI on

	baseline: scVI					all scenarios + real data. Add to Table 1.
3	Missing baseline: ALRA	R1, R3	Critical	No	N/A	Run ALRA (R package or Python port) on all scenarios + real data. Add to Table 1.
4	No ablation study	R1, R2, R3	Critical	No	N/A	Create variants: (a) no biozero regularization ($\lambda_{\text{bio}}=0$), (b) no shrinkage ($\alpha=0$), (c) uniform masking ($p_{\text{zero}}=p_{\text{nz}}$), (d) no noise injection. Report each variant in a new ablation table.
5	No statistical significance / error bars in Table 1	R2	Major	Partially	Fig. 2 shows error bars	Add $\pm$ std to ALL methods in Tables 1 and 2. Consider paired Wilcoxon test across 13 scenarios.
6	No downstream task evaluation	R2, R3	Major	No	N/A	After imputation, run Leiden clustering on the imputed matrix. Compare ARI/NMI against ground-truth cell labels. Also run trajectory inference (e.g., diffusion pseudotime) on trajectory scenarios.
7	Per-scenario results hidden	R2, R3	Major	No	N/A	Add a supplementary table with per-scenario MSE for all methods. This verifies the 12/13 win claim.
8	Hyperparameter sensitivity	R2	Major	No	N/A	Vary $\alpha \in \{0, 0.1, 0.3, 0.5, 1.0\}$, $\lambda_{\text{bio}} \in \{0, 0.5, 2.0, 5.0\}$, $p_{\text{zero}} \in \{0.005, 0.01, 0.02, 0.05\}$. Plot MSE vs. each on one representative scenario.
9	Mathematical error in p_{bio}	R1	Medium-High	No	Sec. 3.5	Fix the numerator from $(1-\delta_{ij})f_{ij}(0)$ to $f_{ij}(0)$. Re-run experiments with corrected formula and report whether

results change.

10	Overclaiming: "scalable"	R3	Major	No	Title, Abstract	Either (a) demonstrate superior scaling (e.g., via batched GPU processing) or (b) remove "Scalable" from the title and temper claims.
11	Overclaiming: cross-modality applicability	R3	Medium	No	Abstract	Change "applicable to" → "potentially applicable to" or remove entirely.
12	Overclaiming: "significantly improves"	R3	Medium	No	Abstract	Remove "significantly" or back it up with statistical tests.
13	gNRMSE > 1 not discussed	R3	Medium	No	N/A	Add a sentence in Sec. 4.1 acknowledging that gNRMSE > 1 indicates room for improvement and contextualizing it (e.g., the metric is strict because of low-variance genes).
14	Shrinkage mechanism ad hoc	R3	Medium	Partially	Sec. 3.7	Add motivation for the functional form. Compare to at least one alternative (e.g., soft-thresholding at a biozero-derived threshold). Ablate over α .
15	Self-citation / ccImpute bias	R3	Low	N/A	N/A	Not fixable, but adding stronger baselines (scVI, ALRA) mitigates the perception.
16	Notation conflicts (d, B, w)	R1	Low	No	Multiple	Use d_lat for bottleneck, d_ij for dropout; B for mini-batch and B_0 for biozero set.
17	Dangling "m" in Sec. 3.4	R1	Minor	No	Sec. 3.4	Remove.
18	"Algorithm 1 (Algorithm 1)"	R1	Minor	No	Sec. 3.9	Remove duplicate.

19	Figure 1 p ⁽⁰⁾ _bio superscript inconsistency	R1	Minor	No	Fig. 1 caption	Unify notation.
20	Figure 2 biozero-MSE y- axis below zero	R2	Minor	No	Fig. 2	Set ylim lower bound to 0.
21	Figure 1 training objective omits L_nz	R1	Minor	No	Fig. 1	Update diagram to show $L = L_{\text{mask}} + \lambda_{\text{bio}} L_{\text{bio}} + \lambda_{\text{nz}} L_{\text{nz}}$.
22	"Baseline" not defined in Table 1	R2	Minor	No	Table 1	Add a footnote or clarifying sentence: "Baseline = observed logcounts with no imputation."
23	GPU hardware not specified	R2	Medium	No	N/A	State the GPU model and confirm all methods used the same hardware.
24	NeurIPS template not used	ALL	Critical	No	Entire paper	Paper uses UAI 2026 template. Must convert to NeurIPS 2026 LaTeX template.
25	No broader impact / limitations section per NeurIPS	R3	Medium	Partially	Sec. 6	Sec. 6 exists but may not meet NeurIPS formatting requirements. Verify compliance with NeurIPS 2026 checklist.

Task 6: Final Venue Compliance Checklist (NeurIPS 2026)

Formatting Compliance

- ☐ **FAIL: Correct LaTeX template used** — Paper uses UAI 2026 template (two-column). NeurIPS uses single-column, different formatting. **MUST CONVERT.**
- ☐ **UNKNOWN: Page limit met** — NeurIPS 2026 typically allows 9 pages of content + unlimited references/appendix. Current paper is ~9 pages in UAI format, which will expand in NeurIPS single-column format. May need trimming.

- ☐ **FAIL: Correct column format** — Paper is two-column (UAI). NeurIPS requires single-column.
- ☐ **FAIL: Font size and margins** — Do not match NeurIPS specs. Need neurips_2026.sty.
- ☒ Figures appear high resolution (vector/matplotlib).
- ☐ **PARTIAL: Figures readable in grayscale** — Figure 2 uses many colors for different methods. Verify distinguishability in grayscale. Figure 3 likely OK with different line styles.
- ☒ Figure captions are mostly self-contained (Figure 1 is detailed, Figures 2–3 are acceptable).
- ☒ Table formatting is clean with clear headers.
- ☒ Algorithm box properly numbered (Algorithm 1).
- ☐ **FAIL: Citation format** — Must match NeurIPS natbib style. Currently uses UAI format.
- ☐ **UNKNOWN: Anonymization** — Paper appears anonymous (no author names visible). Check for self-identifying language. The paper cites ccImpute [Malec et al., 2022] as "Our previous work" — **this is a de-anonymization violation for double-blind review**. Change to "ccImpute [Malec et al., 2022]" without "our previous work."
- ☐ **PARTIAL: Required sections** — Limitations/Ethics section exists (Sec. 6). NeurIPS requires a paper checklist (reproducibility checklist). **Checklist not included.**
- ☐ **FAIL: Appendix/supplementary** — No supplementary material. Per-scenario tables, ablations, and hyperparameter sensitivity should go in a supplementary.

Content Quality

- ☐ **PARTIAL: Abstract** — Too long and discursive. NeurIPS abstracts should be ~150 words. Current abstract is ~200+ words and includes broad claims about other modalities. Tighten to problem→gap→method→result structure.
- ☒ Introduction makes contributions clear within first 2 pages (contribution list on p.2).
- ☐ **FAIL: Every contribution claimed is substantiated** — Contribution bullet 4 claims "benchmark against representative baselines" but the baselines are outdated. The cross-modality applicability claim is unsubstantiated.
- ☐ **PARTIAL: Related work covers last 24 months** — Related work sections cover through 2024 (scMAE/Fang et al., 2024). However, no references from 2025 are included, and several important 2022–2024 methods (scVI, ALRA as baseline) are missing.
- ☒ Method section is precise enough to reimplement from.
- ☐ **FAIL: All notation defined before use** — Some notation conflicts (d, B, w). $p^{(0)}_{\text{bio}}$ in Figure 1 never defined in text.
- ☐ **PARTIAL: All design choices justified** — Many hyperparameters stated without justification (why $\alpha=0.3$? why $v=0.02$? why those architecture widths?).
- ☐ **FAIL: Experiments include current SOTA baselines** — Missing scVI, ALRA, and 2023–2025 methods.
- ☐ **FAIL: Ablation study covers all proposed components** — No ablation study exists.
- ☐ **PARTIAL: Results analysis explains WHY** — Some discussion of why biozero metrics are

higher (trade-off), but no deeper analysis of when/where MaskClass succeeds or fails.

- ✓ Limitations are discussed (Sec. 6).
- ✓ Conclusion (implicitly in Sec. 6) does not introduce new information.

Top-Paper Quality Signals

- **FAIL: Contribution understandable from title + abstract + figures alone** — Figure 1 is good but Figures 2–3 require reading the text. No "hero figure" showing the key insight visually.
- **PARTIAL: Not easily dismissible as "just X plus Y"** — The paper could be dismissed as "masked autoencoder + biozero prior + shrinkage." Needs stronger theoretical or empirical justification for the combination.
- ✓ Every figure serves a purpose (no decorative filler).
- **PARTIAL: One coherent story** — The story is coherent but incomplete without real-data validation.
- **PARTIAL: Writing is dense** — Generally well-written but Sec. 3.1 (simulation details) is very dense and could be moved to supplementary.
- **FAIL: First two pages compelling enough for a skimming reviewer** — The introduction is well-written but the lack of a real-data result on page 1 or 2 will cause a skimming reviewer to look for it, not find it, and prejudge negatively.

End of Review Report.